

Retail Sales, Weather & Demographics Analysis

Miami & Jacksonville Retail Shops, Dashboard Findings & Recommendations

Problem Statement

Daily sales fluctuate, but leadership lacks clarity on why. Understanding how weather and customer demographics impact performance will provide insights to improve planning, staffing, and promotions.

Goal: Provide insights to improve planning, staffing, and promotions.

Our Approach

- **Build a database:** Gather and clean daily sales, weather, and survey data.
- **Develop SQL:** Join datasets on date into one unified analytical view.
- **Analyze data:** Explore descriptive statistics and visual patterns.
- **Create dashboards:** Build Power BI dashboards for stakeholders.

Analysis Objectives

- Link sales data with weather patterns and survey demographics.
- Identify correlations: temperature vs. sales, rainfall effects, weekend lift.
- Compare shop performance and seasonal patterns.
- Build a foundation for dashboards and predictive modeling.

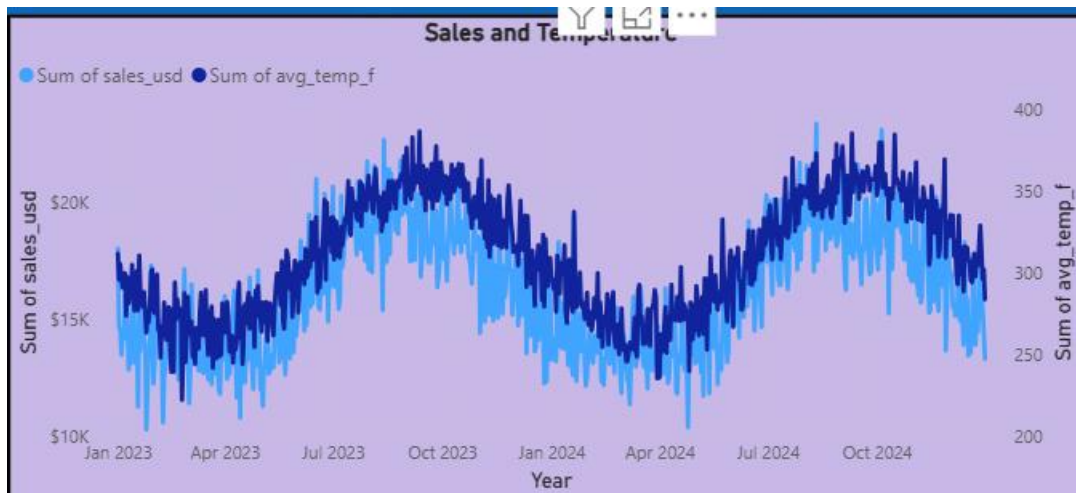
Dashboard Questions, Answers & Next Steps

Finding:

The dashboard shows a clear relationship between temperature and daily sales. Warmer days generally attract more customers and lead to higher revenue. Rain mainly reduces customer visits rather than overall sales, suggesting that existing customers continue to make purchases, while fewer new customers come to the store.

How we can improve this result:

The chart suggests that weather affects sales, but we need statistical testing to confirm it. The next step is to calculate the correlation between `sales_usd` and `avg_temp_f` and compare sales on rainy versus non-rainy days using Python. This will help determine whether the relationship is strong enough to guide staffing and promotional decisions.



Finding:

Looking only at total sales can be misleading because shops with more visitors naturally make more money. A better measure is sales per customer, which shows how much each customer spends on average. This gives a fairer comparison of how well each shop performs.

How we can improve this result:

To better understand why one shop performs better than another, we can analyze sales per customer together with weather and customer information for each shop. For example, the top-performing shop may have better weather or more customers who usually spend more. This helps explain why a shop performs well and provides useful insights that can be applied to other locations.



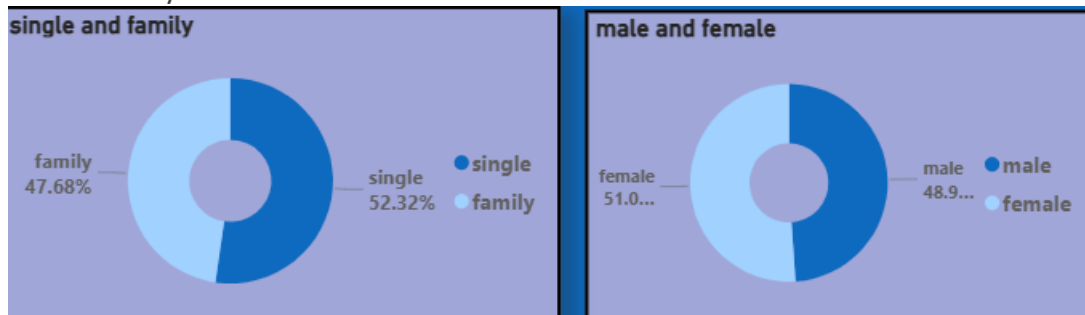
Finding:

The current donut charts give a clear overall snapshot of the customer base composition across the full date range covered by the data.

How we can improve this result:

The donut charts answer 'who are our customers' well, but not yet 'how does this change over time', that part of the question needs a trend visual, not a snapshot. The

recommended next step is to add a stacked bar or area chart showing the family/single and male/female split by month. This would let us see, for example, whether families become a bigger share of customers in summer, which would directly justify family-focused weekend promotions at a specific time of year rather than year-round.



Finding:

The pivot table shows that certain months consistently outperform others, indicating a seasonal rhythm to sales rather than random day-to-day noise.

How we can improve this result:

A pivot table is useful for stakeholders who want to drill into exact numbers, but it doesn't make the seasonal pattern immediately visible at a glance. Adding a proper seasonal line chart (average sales by month, across years if more than one year of data becomes available) would make this pattern something leadership can see in two seconds instead of reading a table. If more historical data is collected going forward, this is also the piece that would feed directly into a demand-forecasting model.

Business Recommendations

- **Increase staffing and inventory on hot summer days** to meet higher customer demand and avoid stock shortages.
- **Offer promotions on rainy days** to encourage customer visits and reduce the impact of lower foot traffic.
- **Focus expansion efforts in Miami during peak tourism seasons**, where demand is strongest and revenue opportunities are highest.
- **Strengthen customer loyalty in Jacksonville** through targeted loyalty programs and personalized offers to increase repeat purchases.
- **Adjust marketing campaigns based on customer segments**, targeting families on weekends and individual customers on weekdays to better match shopping behavior.

Conclusion

The analysis shows that sales are influenced by a combination of weather conditions, store location, and customer demographics. Businesses can improve revenue and operational efficiency by aligning staffing, inventory, promotions, and marketing strategies with these patterns, enabling more informed and data-driven decision-making.